2520

M.Sc. CHEMISTRY (Semester-II)
Examination, 2024
Paper - Sixth
Coordination Chemistry

Duration of Examination: 3 Hours
परीक्षा की अवधि: 3 घण्टा

Max. Marks: 50
पूर्णांक: 50

Instructions to the Candidates:
परीक्षार्थी के लिए निर्देश:-

Note:- The question paper is divided into three parts, Part-A, Part-B and Part-C.

Part-A (Compulsory)

Answer all ten questions (upto 20 words each). Each question carries equal marks. *(Marks-10)*

Part-B (Compulsory)

Answer all five questions (upto 50 words each). Each question carries equal marks. *(Marks-10)*

Part-C

Answer any three questions (upto 400 words each). Selecting one question from each unit. Each question carries equal marks. *(Marks-30)*

Part-A

1- What is $M-L-\pi$ bonding complex?

2- What is chelate effect?

3- Explain natural order of stability or "Irving-William order".

4- Discuss the spectroscopic term symbol of $^{3}F_{2}$.

5- Explain nature of M-L Bond according to CFT.

6- What is spincrossover?

7- Calculate the 10Dq value of $[Ti(H_2O)_6]^{+3}$ complex.

8- What are dinitrogen complexes?

9- Give two example of π bonding polyatomic ligand.

10- Write the example of paramagnetic metal corbonyl complex.

Part-B

11- Explain labile and inert complexes.

12- Write about spectro chemical series.

13- Draw orgel diagram for d^7 configuration in octahedral complexes.

14- Explain vibrational spectra of metal carbonyls for bonding and structure elucidation.

15- Calculate the effective atomic number (EAN) of $Cr(CO)_6$ and $Fe_2(CO)_9$ metal Carbonyl complex.

Part-C

Unit-I

16- Discuss the factors affecting the stability of metal complexes with examples.

OR

Discuss the MOT diagram for octahedral complexes.

(i) When ligand are stronger.

(ii) When ligand are weaker.

Unit-II

17- Discuss the charge transfer spectra in octahedral and tetrahedral complexes.

OR

Write short note on following:-

(a) Anomalous magnetic moments.

(b) Megnetic exchange coupling.

Unit-III

18- Discuss the bonding and structure of metal corbonyl complexes.

OR

Discuss the following :-

(a) Structure of dioxygen complexes.

(b) Important reactions of transition metal nitrosyl complexes.

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M.Sc. CHEMISTRY (Semester-II)
Examination, 2024
Paper-Seventh
Reaction Mechanism - II and Stereochemistry

Max. Marks: 50
पूर्णांक: 50

Duration of Examination: 3 Hours
परीक्षा की अवधि: 3 घण्टा

Instructions to the Candidates:
परीक्षार्थी के लिए निर्देश:-

Note:- The question paper is divided into three parts, Part-A, Part-B and Part-C.

Part-A (Compulsory)

Answer all ten questions (upto 20 words each). Each question carries equal marks. *(Marks-10)*

Part-B (Compulsory)

Answer all five questions (upto 50 words each). Each question carries equal marks. *(Marks-10)*

Part-C

Answer any three questions (upto 400 words each). Selecting one question from each unit. Each question carries equal marks. *(Marks-30)*

Part-A

1- What is wittig reaction?

2- Why is metal hydride reduction preferred over catalytic hydrogenation?

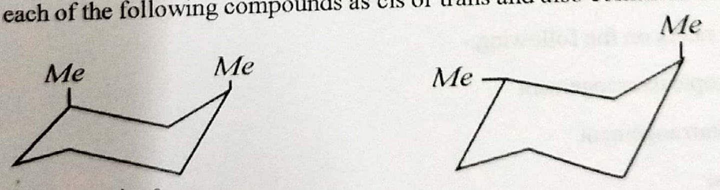
3- Give one example each of chemoselective and regioselective reaction.

4- What do you mean by enantiotopic and diastereotopic atoms?

5- Briefly give the mechanism of Hydroboration.

6- Why Biphenyls exhibit optical activity even in the absence of chiral carbon?

7- Label each of the following compounds as cis or trans and also comment on their chirality.



8- What is Ene reaction?

9- Draw the Frontier Orbitals of Butadiene.

10- Why pericyclic reactions are stereospecific?

Part-B

11- Give the mechanism of ammonolysis of esters.

12- Explain sharpless asymmetric epoxidation.

2521

13- What is the effect of conformation on reactivity?

14- Explain 2+2 addition of ketenes.

15- What are conrotatory and disrotatory motions? Explain giving examples.

Part-C

Unit-I

16- Write short notes on the following:- (3+3+4)

(i) Aldol reaction.

(ii) Mannich reaction

(iii) Hydrolysis of esters and amides.

OR

Explain the hydrogenation of double bond, triple bond and aromatic rings. 10

Unit-II

17- Write short notes on the following:- 5+5

(a) Elements of symmetry

(b) Methods of resolution of enantiomers.

OR

(a) Explain conformational analysis of cycloalkanes. 5

(b) Describe chirality due to helical shape. 5

Unit-III

18- Classify the pericyclic reactions citing one example of each and describe briefly the PMO approach for 4n system. 10

OR

Write short notes on the following:- 2½+2½+2½+2½

(a) 5,5-Sigmatropic rearrangement

(b) Aza-cope rearrangement

(c) Ene reaction

(d) Chelotropic reaction.

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M.Sc. CHEMISTRY (Semester-II)
Examination, 2024
Paper-Eighth (VIII)
Physical Chemistry-II

Duration of Examination: 3 Hours | *Max. Marks: 50*
परीक्षा की अवधि: 3 घण्टा | *पूर्णांक: 50*

Instructions to the Candidates:
परीक्षार्थी के लिए निर्देश:-

Note:- The question paper is divided into three parts, Part-A, Part-B and Part-C.

Part-A (Compulsory)

Answer all ten questions (upto 20 words each). Each question carries equal marks. *(Marks-10)*

Part-B (Compulsory)

Answer all five questions (upto 50 words each). Each question carries equal marks. *(Marks-10)*

Part-C

Answer any three questions (upto 400 words each). Selecting one question from each unit. Each question carries equal marks. *(Marks-30)*

Part-A

1- Define corrosion.

2- Define exchange current density.

3- What is DME in polarography?

4- Write one postulate of B.E.T. equation.

5- Define the term adsorption.

6- Define surface active agents.

7- What are emulsions?

8- Define grand canonical ensembles.

9- Write the expression for fermi dirac statistics.

10- Write postulate of ensembles.

Part-B

11- What are the causes of corrosion of metals?

12- Define liquid crytal polymers?

13- What are macromolecules?

14- Give the expression for internal energy in terms of partition function.

15- Define translational partition function.

Part-C

Unit-I

16- What is the principle of polarography? Explain Ilkovic equation and half wave potential.

OR

Define Ion-solvent interaction and discuss the thermodynamics of electrified interfaces.

Unit-II

17- What is CMC? What are the factors affecting the CMC of surfactants?

OR

Define pressure difference across curved surface and derive laplace equation.

Unit-III

18- Explain thermodynamic probability and derive an expression for most probable distribution.

OR

Explain partition function and derive an expression for equilibrium constant in terms of partition function.

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M.Sc. CHEMISTRY (Semester-II)
Examination, 2024
Paper-IX
Group Theory and Spectroscopy

Duration of Examination: 3 Hours | *Max. Marks: 50*

परीक्षा की अवधि: 3 घण्टा | *पूर्णांक: 50*

Instructions to the Candidates:

परीक्षार्थी के लिए निर्देश:-

Note:- The question paper is divided into three parts, Part-A, Part-B and Part-C.

Part-A (Compulsory)

Answer all ten questions (upto 20 words each). Each question carries equal marks. *(Marks-10)*

Part-B (Compulsory)

Answer all five questions (upto 50 words each). Each question carries equal marks. *(Marks-10)*

Part-C

Answer any three questions (upto 400 words each). Selecting one question from each unit. Each question carries equal marks. *(Marks-30)*

Part-A

1- What are the fundamental symmetry operations?

2- Find out the symmetry element and point groups in H_2O.

3- What do you mean by Rayleigh scattering?

4- What do you understand by emission spectra?

5- Define internal conversion.

6- Give any two surface applications of photo acoustic spectroscopy (PAS).

7- Which of the following systems will show ESR spectra?

8- Predict ESR spectrum of BH_3 radical.

9- What do you mean by zero field splitting?

10- Give any two examples of charge Transfer spectra.

Part-B

11- Discuss point symmetry group.

12- Define mutual exclusion principle.

13- Differentiate between radiative and non radiative decay.

14- Describe the basic principle of photo electron spectroscopy (PES).

15- Discuss spin Hamiltonian.

Part-C

Unit-I

16- Write short notes on the following:-

(a) CARS (coherent Anti Stokes Raman spectroscopy)

(b) Relation between orders of a finite group and its subgroup.

OR

Explain the classical and quantum theories of Raman effect in detail.

Unit-II

17- Write short notes on the following:-

(a) Franck-Condon principle.

(b) Spectra of transition metal complexes.

OR

Describe the Auger Electron spectroscopy (AES) in detail.

Unit-III

18- Write short notes on the following:-

(a) Factors affecting the 'g' value.

(b) Spin densities and Mcconnell relationship.

OR

What are isotropic and anisotropic hyperfine coupling constants?

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